Topic 10: Equilibrium I

In order to develop their practical skills, students should be encouraged to carry out a range of practical experiments related to this topic. Possible experiments include investigating equilibrium systems, such as iron(III) – thiocyanate, or the effect of temperature on the equilibrium between $[Co(H_2O)_6]^{2+}$ and $[CoCl_4]^{2-}$

Mathematical skills that could be developed in this topic include deriving an algebraic expression for the equilibrium constant.

Within this topic, students can consider how an appreciation of equilibrium processes, coupled with kinetics, can lead chemists to redevelop manufacturing processes to make them more efficient.

Students should:
1. know that many reactions are readily reversible and that they can reach a state of dynamic equilibrium in which: i the rate of the forward reaction is equal to the rate of the backward reaction ii the concentrations of reactants and products remain constant
2. be able to predict and justify the qualitative effect of a change in temperature, concentration or pressure on a homogeneous system in equilibrium
3. evaluate data to explain the necessity, for many industrial processes, to reach a compromise between the yield and the rate of reaction
4. be able to deduce an expression for K_c , for homogeneous and heterogeneous systems, in terms of equilibrium concentrations